

IEEE BIBM 2026 — Doctoral Forum Cover Note

Dear Dr. Yi He and Dr. Xingquan Zhu,

My name is Andrew A. Lane, and I am a doctoral researcher in Electrical and Computer Engineering at Stony Brook University working at the intersection of neuromorphic computing, biomedical signal analysis, and affective EEG/ERP representation learning. I am pleased to submit the attached manuscript, “Spiking Reservoir State-Space Encoding for Perturbation-Characterized Affective EEG Signal Analysis,” for consideration in the IEEE BIBM 2026 Doctoral Forum. The paper evaluates a fixed leaky integrate-and-fire (LIF) reservoir with a six-bin spike-count (BSC6) code as a nonlinear temporal encoding operator for affective EEG/ERP observations. Using subject-grouped validation against a classical bandpower/ERP baseline, the study reports balanced accuracy, macro-F1, and macro one-vs-rest AUC, and characterizes the encoding under additive amplitude noise, temporal jitter, and channel dropout, separating representation-level corruption from full raw-signal recomputation.

The result is deliberately bounded. The reservoir embedding is not presented as a uniformly stronger static classifier than classical ERP-derived features, but it exhibits a distinct perturbation-response profile: it retains balanced accuracy more strongly under representation-level amplitude noise and channel dropout, while remaining sensitive to temporal misalignment and severe channel loss when the representation is recomputed from corrupted signals. Characterizing what temporal information a spiking representation preserves, transforms, or suppresses, rather than reporting a single clean-accuracy number, gives a reproducible operating-regime account. By characterizing how a temporal reservoir representation changes under amplitude noise, temporal jitter, and channel dropout, the study supports more auditable biomedical signal-analysis pipelines for EEG/ERP data before downstream clinical interpretation is considered.

We believe the work fits the IEEE BIBM Doctoral Forum. Its contribution is original and relevant to the BIBM community: it reframes a spiking reservoir as a perturbation-characterized temporal measurement operator for affective EEG/ERP rather than as another endpoint classifier. The methodology and evaluation are designed for soundness and clarity: subject-grouped validation, a classical EEG/ERP baseline, balanced accuracy, macro-F1, macro one-vs-rest AUC, and a fold-local PCA sensitivity analysis that bounds the transductive-projection concern. The problem it addresses, representational stability of biomedical signals under controlled corruption, is a significant and underreported question for reproducible EEG/ERP analysis. As student-led doctoral-stage research, the first author is a doctoral student in Electrical and Computer Engineering at Stony Brook University, the primary contributor, and the intended presenter; a one-page CV of the first author is appended as required by the call. This submission forms one component of a broader doctoral framework for neuromorphic signal processing of affective neural data and is intentionally focused on the reservoir temporal-encoding component; it does not duplicate the broader journal manuscript under preparation.

Sincerely,

Andrew Lane